inherited-f4-next and fes-f2 end to end (2026-11-12)

`inherited-f4-next`: the same counted work with faster kernels, and `fes-f2` end to end (2026-11-12)

Owner's ask (2026-11-12): "Speed it all up". The M4GB side is `../m4gb_update_signatures_20261112/`. This study covers the in-house engines that the algebraic index-calculus oracle actually runs.

Correction to the brief

The change was briefed as "`inherited-f4` is the best engine on every cell". That holds for end-to-end S in the e2e ladder (`../m4gb_e2e_ic_ladder_20261110/`), which prices `inherited-f4` from its word-XOR count, and in that ladder `fes-f2` was not an arm. On wall time per call, `fes-f2` (Gray-code exhaustive search) is below `inherited-f4` up to 24 unknowns, here and already in the descent ladder (`../m4gb_descent_ladder_20261110/RESULTS.md`, R 21: 3.9 against 9.5 ms) `[pcc:unverified]`. `fes-f2` is also priced in word XORs (two per Gray-code point), so its S is on the same basis as `inherited-f4`'s; at 14 unknowns its count per call is below `inherited-f4`'s (about 33 thousand against 49 thousand in the profiling runs), from 20 unknowns up it is 4 to 13× above. The end-to-end run below measures what that does to S.

Profile (Callgrind, `inherited-f4`, AVX2 path)

`gf2_elim` chooses AVX-512 at run time on this host; Valgrind cannot decode it, so the profile is of the AVX2 path (`KIC_GF2_FORCE_AVX2=1`), instruction counts with cold caches `[pcc:unverified]`:

inclusive share	**K 23:12:2**	**R 21:11:2**	**K 13:7:2**
tree specialisation (`specialise_shared`)	64 %	59 %	55 %
↳ eager rewrite of kept rows (`apply_step_bmi2`)	20 %	16 %	11 %

inclusive share	K 23:12:2	R 21:11:2	K 13:7:2
↳ re-reduction of displaced rows (`ReducedBasis::insert`)	16 %	13 %	14 %
root (`reduce_inherited`)	22 %	29 %	31 %
↳ Macaulay build	12 %	19 %	20 %
↳ root echelon (`eliminate_core`)	9 %	9 %	9 %
system substitution	9 %	5 %	5 %

The structure-reuse-across-targets question (“dynamic programming”): the column layout is already cached across targets (keyed by the occurring variables and the degree); what a root still does per target is numeric, and 55 to 64 % of the cost is the specialisation down the tree, which is counted work. A complete-layout formula index was tried and gave nothing measurable; the existing Gröbner-trace option (`KIC_F4_ROOT_REPLAY`) drops rows and can change the tree, so it does not meet the identical-decisions rule.

What changed (engine `inherited-f4-next`; `inherited-f4` unchanged)

A per-thread switch (`koblitz_groebner::with_next_kernels`) selects, for the same eliminations and rewrites:

1. SIMD row XOR in `ReducedBasis::insert` (AVX-512 or AVX2 at run time);
2. the `bitop/v1` tally counted inside the rewrite pass instead of a second pass over the same words (identical tally);
3. Four Russians table reuse (`gf2_elim::set_reuse_table_thread`, the exact `KIC_GF2_REUSE_TABLE` behaviour);
4. existing exact matrix builders that were off by default (direct fused pack, support-local streaming and bitmap columns), whose tests assert identical matrices;
5. environment variables read once instead of per node.

None of these changes the engine’s `ops` (word XORs): the same work, done cheaper. Tests: `koblitz_groebner::tests::next_kernels_leave_decisions_and_word_xors_unchanged` (33 fixed systems, cold and warm, identical roots, tree statistics and word count), `gf2_elim::tests::reuse_table_override_leaves_rows_and_count_unchanged`, `inherited_f4::tests::simd_row_xor_matches_the_scalar_one`; module suites `koblitz_groebner` (92), `gf2_elim` (12), `inherited_f4` (14), `ic_framework::solvers` (13), `pq_f4_f2` (13) pass. Written by a subagent to a frozen brief; reviewed in the main session.

Run 1: paired per call (`run_ab.sh`)

`ic descent --solver inherited-f4 --solver inherited-f4-next --solver fes-f2`, Koblitz and random cells, m = 2 at 14 to 24 unknowns plus Koblitz 30 and 32 (degrees 25 and 27 are composite and have no Koblitz cell), 6 seeded targets × 3 repeats,

60 s per call, one thread; engine order rotates per repetition. Record `20261112T0630Z-inherited-f4-next-ab`.

Expectation: next/old 0.85 to 0.95 per call on every cell, `ops` equal on every target, 0 disagreements; `fes-f2` below both on wall up to 24 unknowns and above them at 30 to 32. Falsifier: any `ops` difference or disagreement, or a median paired ratio at or above 1.

Run 2: end to end (`run_e2e.sh`)

`ic bench --sweep` on the six instances of the e2e ladder (Koblitz 13, 17, 19, 23; random 15, 21; same subspace bases, 2 repeats, 60 s per call, 16 matched rho runs), the algebraic oracle with `inherited-f4`, `inherited-f4-next` and `fes-f2`, the pair table as reference row.

Expectation, stated before the run: `inherited-f4` and `inherited-f4-next` have the same S on every instance (same count, pinned pricing) and the second has the lower wall; `fes-f2` has a lower S than `inherited-f4` at degree 13 (14 unknowns) and a higher S from degree 17 or 19 up. If the first holds, it is a measured gain `[pcc:unverified]` on the best algebraic row at degree 13 and nothing else. Falsifier: `inherited-f4` and `-next` S differing beyond run noise, or `fes-f2` S at degree 13 not below `inherited-f4`'s.

Results

Run 1 `[pcc:unverified]`

2026-11-12 06:30 to 06:32 Z, record `benchmarks/records/20261112T0630Z-inherited-f4-next-ab.json` (measured; raw `results.tar.gz`, `results_ab.json`). Full table `RESULTS_AB.md`, graph `results_ab.svg/png`.

unknowns	`inherited-f4` ms K / R	`-next` ms K / R	paired next/old	`fes-f2` ms K / R	word XORs `inherited-f4` / `fes-f2` (K)
14	1.03 / 1.19	0.94 / 1.13	0.92 / 0.93	0.09 / 0.10	49 k / 33 k
16	2.40 / 1.30	2.22 / 1.18	0.93 / 0.91	0.19 / 0.12	124 k / 132 k
18	2.21 / 2.22	2.08 / 2.04	0.94 / 0.92	0.24 / 0.25	251 k / 525 k
20	4.25 / 5.58	3.65 / 5.11	0.86 / 0.89	0.82 / 1.24	528 k / 2.10 M
22	— / 7.23	— / 6.38	0.88	— / 2.77	1.06 M / 8.39 M (R)
24	15.7 / 14.3	13.7 / 12.5	0.89 / 0.89	10.7 / 11.3	2.50 M / 33.6 M
30 (K)	90.5	81.7	0.89	705	20.0 M / 2.15 G
32 (K)	217	192	0.87	3,036	42.9 M / 8.59 G

- `inherited-f4-next` is 0.86 to 0.94× of `inherited-f4` per call on every cell, below it on 74 of 78 targets, with the same per-target `ops` on every target and repeat and 0 disagreements with `fes-f2`. The expectation held; the falsifier did not fire.
- `fes-f2` is below both inherited engines on wall up to 24 unknowns (0.08 to 0.11× at 14 to 18, 0.7 to 0.8× at 24) and 7.8 and 14× above at 30 and 32. Its word-XOR count is below `inherited-f4`'s only at 14 unknowns.
- The `ops` are unchanged, so the S of an `inherited-f4` row priced from its pinned word-XOR count does not move; the change lowers wall time only.

Run 2 `[pcc:unverified]`

2026-11-12 06:33 to 06:36 Z, record `benchmarks/records/20261112T0633Z-inherited-f4-next-e2e.json` (measured; raw `results_e2e.tar.gz`, `results_e2e.json`). Full table `RESULTS_E2E.md`, graph `results_e2e.svg/png`. Every one of the 24 configurations recovered its planted logarithm in every repeat, and so did every matched rho run.

instance (unknowns)	rho S	pair table S	`inherited-f4` S	`-next` S	`fes-f2` S	ms per call `inherited-f4` / `-next` / `fes-f2`
Koblitz 13 (14)	2.76	197.7	123.1 (117.8–128.4)	123.1	**94.7 (83.7–105.7)**	0.76 / 0.74 / 0.06
random 15 (16)	2.64	474.0	357.5	357.5	426.5	1.91 / 1.41 / 0.18
Koblitz 17 (18)	2.02	493.0	249.1	249.1	494.0	2.80 / 2.78 / 0.30
Koblitz 19 (20)	1.48	1,063.2	411.0	411.0	2,548.0	4.57 / 4.12 / 0.73
random 21 (22)	1.36	1,975.8	2,758.3	2,758.3	21,807.9	8.83 / 7.70 / 3.13
Koblitz 23 (24)	0.92	3,986.2	1,852.5	1,852.5	18,444.0	16.14 / 14.82 / 12.00

S = total group-addition equivalents / sqrt(r), mean over 2 repeats (range for degree 13). Every algebraic row is priced from its solver's word-XOR count (`pinned` on the Koblitz instances, `measured` ratio on the random ones), the same basis for all three engines.

- **Degree 13: `fes-f2` lowers the best index-calculus row from 123.1 to 94.7 (0.77×), 34.3× the matched rho instead of 44.6×.** The two repeats of each engine do not overlap (83.7–105.7 against 117.8–128.4). At 14 unknowns, two word XORs per Gray-code point over 2^14 points cost less than the inherited tree's eliminations. This is the expectation stated before the run; the falsifier did not fire. The distance remaining to rho at degree 13 is 34.3×.
- **From 16 unknowns up `fes-f2` has the higher S** (1.2× at random 15, 2× at Koblitz 17, 6 to 10× from 19 up): its count doubles per unknown while the inherited tree's grows far more slowly. `inherited-f4` stays the best algebraic row there, and the pair table stays best on random 21.

- **`inherited-f4-next` has exactly the S of `inherited-f4` on every instance** (same counted work) and a lower wall per call inside the whole pipeline: 0.74 at random 15, 0.87 at random 21, 0.90 at Koblitz 19, 0.92 at 23, 0.97 to 0.99 at 13 and 17.
- `fes-f2` tried 47 targets for 31 relations at degree 13 against 42 for 28 with the inherited engines (hit rate 0.657 against 0.654); S counts all of them.

Not a leaderboard row: the row has no replay certificate or validator report (validation gate), the run is local and unisolated, and 2 repeats is a small sample. The next step for degree 13 is a validated isolab run of the `fes-f2` configuration; a size rule in the oracle (`fes-f2` up to 14 unknowns, `inherited-f4` above) would carry the gain into any sweep without choosing the engine per instance.

Sources checked

KB (`research-kb`) unavailable in this session. `docs/ic/FRAMEWORK.md` (engine table: `fes-f2` and `inherited-f4` both priced in word XORs), `research/m4gb_e2e_ic_ladder_20261110/README.md` and its sweeps, `research/m4gb_descent_ladder_20261110/RESULTS.md`, `docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`; the subagent's Callgrind files.

Median over targets of the per-target median wall (ms per call, one thread). Paired: median (min–max) over targets of the per-target ratio. `ops` = median per-target word-XOR count (the engines' counted unit). `dec` = targets decided / targets; every decided answer is checked against the reference `fes-f2`.

family	n	vars	`inherited-f4` ms	`-next` ms	next/old paired	targets next < old	ops equal	`fes-f2` ms	fes/old
K	13	14	1.03	0.94	0.918 (0.90–1.00)	6/6	yes	0.09	0.09
R	13	14	1.19	1.13	0.926 (0.87–0.95)	6/6	yes	0.10	0.08
K	15	16	2.40	2.22	0.930 (0.83–0.95)	6/6	yes	0.19	0.08
R	15	16	1.30	1.18	0.914 (0.90–0.93)	6/6	yes	0.12	0.10
K	17	18	2.21	2.08	0.941 (0.91–1.02)	5/6	yes	0.24	0.11
R	17	18	2.22	2.04	0.920 (0.89–0.97)	6/6	yes	0.25	0.11
K	19	20	4.25	3.65	0.860 (0.81–0.86)	6/6	yes	0.82	0.21

family	n	vars	inherited-f4 ms	-next ms	next/old paired	targets next < old	ops equal	fes-f2 ms	fes/old
R	19	20	5.58	5.11	0.893 (0.84–0.98)	6/6	yes	1.24	0.21
R	21	22	7.23	6.38	0.882 (0.84–0.89)	6/6	yes	2.77	0.36
K	23	24	15.74	13.71	0.886 (0.81–0.90)	6/6	yes	10.73	0.69
R	23	24	14.31	12.47	0.894 (0.85–1.28)	5/6	yes	11.34	0.80
K	29	30	90.47	81.70	0.893 (0.80–0.95)	6/6	yes	705	7.75
K	31	32	217	192	0.872 (0.80–1.15)	4/6	yes	3,036	14.22

Per instance: the matched counted rho (mean S over its runs) and every configuration run end to end. `S` = total group-addition equivalents / sqrt(r), mean over repeats (min–max); `S / rho` = mean s_over_rho; `ok` = every repeat recovered the planted logarithm; `hit` = decomposition hit rate; `tried` = targets tried; `solver` = engine, priced_by, mean calls, budget hits, share of the row's GAE, wall per call.

instance (r)	configuration	S mean (min–max)	S / rh
icv1-f2m15-tm249-9e24ae34 (r = 5503)	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts`, 16 runs	2.64	1.0
	`binary-subspace[dimension=8] + mitm[m=2] + – + walk`	474.05 (473.30–474.80)	179.7
	`binary-subspace[dimension=8] + descent-algebraic[m=2] + inherited-f4 + walk`	357.54 (242.85–472.24)	135.5
	`binary-subspace[dimension=8] + descent-algebraic[m=2] + inherited-f4-next +`	357.54 (242.85–	135.5

instance (r)	configuration	S mean (min–max)	S / rh
	`walk`	472.24)	
	`binary-subspace[dimension=8] + descent-algebraic[m=2] + fes-f2 + walk`	426.51 (289.74–563.28)	161.7
`icv1-f2m21-tm1325-b6925563` (r = 1049239)	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts`, 16 runs	1.36	1.0
	`binary-subspace[dimension=11] + mitm[m=2] + – + walk`	1975.75 (1975.75–1975.75)	1449.0
	`binary-subspace[dimension=11] + descent-algebraic[m=2] + inherited-f4 + walk`	2758.31 (2723.40–2793.21)	2022.9
	`binary-subspace[dimension=11] + descent-algebraic[m=2] + inherited-f4-next + walk`	2758.31 (2723.40–2793.21)	2022.9
	`binary-subspace[dimension=11] + descent-algebraic[m=2] + fes-f2 + walk`	21807.90 (20967.80–22647.99)	15993.8
`icv1-f2m13-t181-`	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by`	2.76	1.0

instance (r)	configuration	S mean (min–max)	S / rh
515ee569 (r = 2003)	doubling, distinguished points, stride starts, 16 runs		
	binary-subspace[dimension=7] + mitm[m=2] + – + walk	197.66 (197.66–197.66)	71.5
	binary-subspace[dimension=7] + descent-algebraic[m=2] + inherited-f4 + walk	123.11 (117.79–128.43)	44.5
	binary-subspace[dimension=7] + descent-algebraic[m=2] + inherited-f4-next + walk	123.11 (117.79–128.43)	44.5
	binary-subspace[dimension=7] + descent-algebraic[m=2] + fes-f2 + walk	94.70 (83.71–105.69)	34.3
icv1-f2m17-tm101-00378d4e (r = 65587)	**matched rho** negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts, 16 runs	2.02	1.0
	binary-subspace[dimension=9] + mitm[m=2] + – + walk	493.04 (492.55–493.54)	244.4
	binary-subspace[dimension=9] + descent-algebraic[m=2] + inherited-f4 + walk	249.09 (102.55–395.63)	123.4
	binary-subspace[dimension=9] + descent-algebraic[m=2] + inherited-f4-next + walk	249.09 (102.55–395.63)	123.4

instance (r)	configuration	S mean (min–max)	S / rh
	`binary-subspace[dimension=9] + descent-algebraic[m=2] + fes-f2 + walk`	493.97 (200.24–787.71)	244.9
`icv1-f2m19-tm797-9c54981b` (r = 262543)	**matched rho** `negation-map r-adding walk: look-ahead, short-cycle escape by doubling, distinguished points, stride starts`, 16 runs	1.48	1.0
	`binary-subspace[dimension=10] + mitm[m=2] + – + walk`	1063.15 (1062.78–1063.52)	719.2
	`binary-subspace[dimension=10] + descent-algebraic[m=2] + inherited-f4 + walk`	411.01 (227.52–594.50)	278.0
	`binary-subspace[dimension=10] + descent-algebraic[m=2] + inherited-f4-next + walk`	411.01 (227.52–594.50)	278.0
	`binary-subspace[dimension=10] + descent-algebraic[m=2] + fes-f2 + walk`	2547.95 (2010.28–3085.63)	1723.6
`icv1-f2m23-tm5197-1f85e9e1` (r = 4196903)	**matched rho** `signed-Frobenius r-adding walk, distinguished points (koblitz_signed_frobenius_rho_reference)`, 16 runs	0.92	1.0
	`binary-subspace[dimension=12] + mitm[m=2] + – + walk`	3986.20 (3986.00–3986.40)	4334.1
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + inherited-f4 + walk`	1852.45 (1498.17–2206.73)	2014.1

instance (r)	configuration	S mean (min–max)	S / rh
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + inherited-f4-next + walk`	1852.45 (1498.17–2206.73)	2014.1
	`binary-subspace[dimension=12] + descent-algebraic[m=2] + fes-f2 + walk`	18444.03 (17128.04–19760.03)	20053.8